Example 4.10

Find the value of $\tan^{-1}(-1)+\cos^{-1}\left(\frac{1}{2}\right)+\sin^{-1}\left(-\frac{1}{2}\right)$.

Solution

Let $\tan^{-1}(-1)=y$. Then, $\tan y=-1=-\tan\frac{\pi}{4}=\tan\left(-\frac{\pi}{4}\right)$.

As $-\frac{\pi}{4}\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, $\tan^{-1}(-1)=-\frac{\pi}{4}$.

Now, $\cos^{-1}\left(\frac{1}{2}\right)=y$ implies $\cos y=\frac{1}{2}=\cos\frac{\pi}{3}$.

As $\frac{\pi}{3}\in[0,\pi]$, $\cos^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{3}$.

Now, $\sin^{-1}\left(-\frac{1}{2}\right)=y$ implies $\sin y=-\frac{1}{2}=\sin\left(-\frac{\pi}{6}\right)$.

As $-\frac{\pi}{6}\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$, $\sin^{-1}\left(-\frac{1}{2}\right)=-\frac{\pi}{6}$.

Therefore, $\tan^{-1}(-1)+\cos^{-1}\left(\frac{1}{2}\right)+\sin^{-1}\left(-\frac{1}{2}\right)=-\frac{\pi}{4}+\frac{\pi}{3}-\frac{\pi}{6}=-\frac{\pi}{12}$.

Example 4.11

Prove that $\tan\left(\sin^{-1}x\right)=\frac{x}{\sqrt{1-x^2}}$, $-1<x<1$.

Solution

If $x=0$, then both sides are equal to 0. ... (1)

Assume that $0<x<1$.

Let $\theta=\sin^{-1}x$. Then $0<\theta<\frac{\pi}{2}$. Now, $\sin\theta=\frac{x}{1}$ gives $\tan\theta=\frac{x}{\sqrt{1-x^2}}$.

Hence, $\tan\left(\sin^{-1}x\right)=\frac{x}{\sqrt{1-x^2}}$. ... (2)

Assume that $-1<x<0$. Then, $\theta=\sin^{-1}x$ gives $-\frac{\pi}{2}<\theta<0$. Now, $\sin\theta=\frac{x}{1}$ gives $\tan\theta=\frac{x}{\sqrt{1-x^2}}$.

In this case also, $\tan\left(\sin^{-1}x\right)=\frac{x}{\sqrt{1-x^2}}$. ... (3)

Equations (1), (2) and (3) establish that $\tan\left(\sin^{-1}x\right)=\frac{x}{\sqrt{1-x^2}}$, $-1<x<1$.

EXERCISE 4.3

1. Find the domain of the following functions :

   (i) $\tan^{-1}\left(\sqrt{9-x^2}\right)$ (ii) $\frac{1}{2}\tan^{-1}\left(1-x^2\right)-\frac{\pi}{4}$.

2. Find the value of (i) $\tan^{-1}\left(\tan\frac{5\pi}{4}\right)$ (ii) $\tan^{-1}\left(\tan\left(-\frac{\pi}{6}\right)\right)$.

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